DAY-3

- **3.1Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.**

Feature	**DDR3**	**DDR4**	**DDR5**
Release Year	2007	2014	2020
Speed (Data Rate)	800–2133 MT/s	1600–3200 MT/s	4800–8400+ MT/s
Operating Voltage	1.5 V (1.35 V for DDR3L)	1.2 V	1.1 V
Performance	Good for older systems	Faster and more efficient than DDR3	Fastest with highest bandwidth and efficiency
Power Consumption	Highest	Lower than DDR3	Lowest, making it more power-efficient

3.2 Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully.

- Shut down your computer completely.
- Disconnect the power cable (and remove the battery if your laptop has a removable one).
- Press and hold the power button for 10–15 seconds to discharge any remaining electricity.
- Open the computer case or laptop back cover.
- Locate the RAM module(s) on the motherboard.
- Take a clear photo showing the installed RAM module.
- Push the retaining clips on both sides of the RAM slot outward.
- Carefully remove the RAM by holding it only by the edges (avoid touching the gold contacts).
- Reinsert the RAM at the correct angle, aligning the notch with the slot.
- Press down firmly until both retaining clips click into place.
- Close the case, reconnect the power, and turn on the computer.
- Verify that the system boots successfully.

3.3 List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

Feature	**HDD (Hard Disk Drive)**	**SSD (Solid State Drive)**	**NVMe SSD**
Speed	Slow (typically 80–160 MB/s)	Fast (typically 500–550 MB/s)	Very Fast (2,000–7,000+ MB/s)
Technology	Spinning magnetic disks	Flash memory (SATA interface)	Flash memory (PCIe interface)
Best For	Large file storage and backups	Everyday computing and faster boot times	Gaming, video editing, and high-performance tasks